# Deathspire Codex: an Immersive Card Game

## Concept

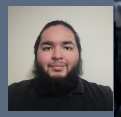
**Your cards dont just hit the table, they enter your world.**

*Deathspire Codex* is an immersive VR card battle game where every card you play becomes a full-scale creature, warrior, or spell that erupts onto the battlefield before your eyes. Command colossal mosnters, cast destructive spells, and duel your opponent In real time as your armies clash around you.

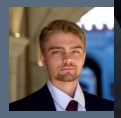
Strategy meets spectacle in this living, breathing fantasy arena that merges the tactical depth of trading card games with the adrenaline of first-person VR combat.

## Meet the Team

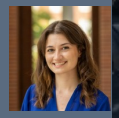

- **Yuchen Xing (Computer Science)**
  - yuchenxing373@gmail.com
  - Shuffling mechanics, AI systems & animations for monsters, networking
- **Albert Castrejon (Computer Science)**
  - albert.castrejon@vanderbilt.edu
  - Menu logic, environement, and theme music; core spawning, audio management, and spell casting systems, mana system, deck org. data structure



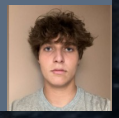
- **Christian Caamano (Computer Science)**
  - christian.d.caamano@vanderbilt.edu
  - Book model, arena environment, player model and inverse kinematics rig, battle music



- **Natalie Dimov (CS/EM)**
  - natalie.m.dimov@vanderbilt.edu
  - Book-card snapping system, worked on developing balanced deck, made a tutorial, worked on gameflow prior to joining arena



- **Nico Davidson (H&OD)**
  - nico.e.davidson@vanderbilt.edu
  - Poster designer, card balancing and ability conceptualizer, and card graphic design.




## Design & User Expirience

**UI User Flows**

Players begin their battling expirience in the **main menu room,** where they must approach the **Deathspire Codex** and open it. From there, players can select **"Host" or "Join"** and then select their deck (**Atk, Def, or Nuetral**). They are then entered into an arena with another player.




Once entered into the arena, players must take cards (entities, spells, or weapons) from the codex in their left hand, and throw them onto the battlefield with their right



*To the left is the moonlit castle-inspired battle arena in all its glory.* Shining under an ominous glow, its the perfect place for an epic battle










Pictured above are 3 cards of different types, from left to right:
**Fireball (Spell), Sword (Weapon), Knight (Entity)**
*Mana cost pictured in the blue droplet at the top right of card*
All cards are cast by dragging them from the codex and throwing them into the arena



*Card entities in battle:*

The **Knight**, **Wraith**, and **Golem** duking it out within the battle arena.

## Challenges

**We faced some signifcant challenges along the way, all requiring their own unique solutions:**

- **Player Rig & Networking Integration**

One of the most technically demanding challenges we encountered was getting the **player rig** (including full body IK, headset tracking, and controller inputs) to function properly within the network multiplayer envirionment. We only overcame this challenge after diving into the network scripts and extensive trial and error with two active headsets to isolate which issues were network vs IK related.

- **Prefab & Unity Wokflow Issues**

Another major challenge we faced involved Unity's prefab workflow, particularly with working collaboratively through version control. Unity rewrites assets files when they are modified, even during minor changes. This made the system prone to merge conflicts that were extremely difficult to resolve. We created temp duplicates of prefabs when changing things to avoid these conflicts

## The Future

**The world of Deathspire Codex is just getting started.**

The future of Deathspire Codex looks like:

- A major expansion of the card library with elemental synergies and abilities adding another level of strategy to the game
- Larger multiplayer modes beyond 1v1 battles (like 1v1v1, or 2v2), leading to more dynamic battles
- Different summoner classes, giving boosts to certain card types based on class
- AR (augmented reality) modes, allowing players to play in their real-life surroundings, turning anywhere into an epic battle arena